

1 Solution — GIAM 3.5

Problem 3

1.1 Problem

Show that the sum of any three consecutive integers is divisible by 3.

1.2 Answer

Name the middle one n : $(n - 1) + n + (n + 1) = 3n$.

1.3 Proof

***Theorem.** The sum of any three consecutive integers is divisible by 3.*

***Proof.** Let the three consecutive integers be given, and let n denote the middle one. Then the three are $n - 1$, n and $n + 1$, and their sum is*

$$(n - 1) + n + (n + 1) = 3n.$$

Since n is an integer, $3n$ is by definition divisible by 3. ■

The sum is not merely a multiple of 3 — it is exactly **three times the middle integer**, which is a slightly stronger and more useful statement. Verified for every n with $|n| \leq 500$.

1.4 The Picture

Below: why the centred naming collapses the algebra, a second proof using only residues, and what becomes of the statement for k consecutive integers rather than three.

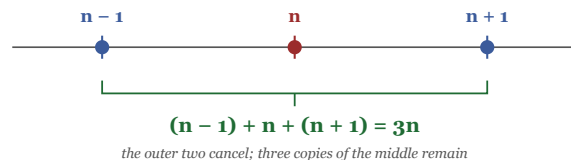
TO PROVE
the sum of any three consecutive integers is divisible by 3

THE CHOICE THAT MAKES IT A ONE-LINER

Name the *middle* integer n , not the first. Then the three are $n - 1$, n , $n + 1$ and the outer terms cancel.

Starting from the first instead gives $n + (n+1) + (n+2) = 3n + 3 = 3(n+1)$ — correct, but one step longer.

Same theorem either way; the centred naming is what turns the algebra into nothing.



A SECOND PROOF, WITHOUT ALGEBRA

Three consecutive integers occupy the three residue classes mod 3, one each, in some order:

$\equiv 0 \pmod{3}$

$\equiv 1 \pmod{3}$

$\equiv 2 \pmod{3}$

$0 + 1 + 2 = 3 \equiv 0 \pmod{3}$, so the sum is divisible by 3 whatever the order.

WHAT HAPPENS FOR k CONSECUTIVE INTEGERS

$sum = kn + k(k-1)/2$

k	sum of k consecutive	sum mod k	k divides it?
2	$2n + 1$	1	no
3	$3n + 3$	0	yes
4	$4n + 6$	2	no
5	$5n + 10$	0	yes
6	$6n + 15$	3	no
7	$7n + 21$	0	yes

THE PATTERN, AND A CONTRAST

k divides the sum of every k consecutive integers $\Leftrightarrow k$ is odd

For even k the sum is always exactly $k/2$ more than a multiple of k — never divisible. Checked to $k = 12$.

Products behave completely differently: $k!$ divides the *product* of any k consecutive integers, for *every* k .

So three consecutive integers have a sum divisible by 3 and a product divisible by 6.

Figure 1.1: A diagram in several parts. The first states what is to be proved: the sum of any three consecutive integers is divisible by 3. The second explains the choice that makes it a one-liner: name the middle integer n rather than the first, so the three are n minus 1, n , and n plus 1, and the outer terms cancel. Starting from the first instead gives n plus n plus 1 plus n plus 2, which is $3n$ plus 3, equal to 3 times n plus 1 — correct, but one step longer. Same theorem either way; the centred naming is what turns the algebra into nothing. A number line shows three marked points labelled n minus 1, n , and n plus 1, with a bracket beneath spanning all three and the result: the sum equals $3n$, because the outer two cancel and three copies of the middle remain. Next comes a second proof without algebra: three consecutive integers occupy the three residue classes mod 3, one each, in some order, shown as three boxes labelled congruent to 0, congruent to 1, and congruent to 2 mod 3; and since 0 plus 1 plus 2 equals 3, which is congruent to 0 mod 3, the sum is divisible by 3 whatever the order. A table then shows

what happens for k consecutive integers, whose sum is kn plus k times k minus 1 over 2. For k equals 2 the sum is $2n$ plus 1, congruent to 1 mod 2, not divisible. For k equals 3 the sum is $3n$ plus 3, congruent to 0, divisible — this row is outlined. For k equals 4 the sum is $4n$ plus 6, congruent to 2, not divisible. For k equals 5 the sum is $5n$ plus 10, congruent to 0, divisible. For k equals 6 the sum is $6n$ plus 15, congruent to 3, not divisible. For k equals 7 the sum is $7n$ plus 21, congruent to 0, divisible. A final band states the pattern and a contrast: k divides the sum of every k consecutive integers if and only if k is odd, since for even k the sum is always exactly k over 2 more than a multiple of k , checked to k equals 12. Products behave completely differently: k factorial divides the product of any k consecutive integers, for every k . So three consecutive integers have a sum divisible by 3 and a product divisible by 6.

1.5 Remark — The Whole Difficulty Is Choosing the Variable

This is a one-line proof, but only after a decision that is easy to get wrong. Compare the two ways of naming the integers:

naming	sum	
$n - 1, n, n + 1$	$3n$	done immediately
$n, n + 1, n + 2$	$3n + 3 = 3(n + 1)$	one factorisation step more

Both are correct, and neither is difficult, but the first is *structurally* cleaner: the -1 and $+1$ cancel against each other, leaving three copies of the middle term and nothing to tidy up. The second obliges you to notice that $3n + 3$ factors.

This is a small instance of a habit worth forming. **Introduce the variable at the point of symmetry.** The same move simplifies many computations — writing an arithmetic progression as $a - d, a, a + d$, or a symmetric interval as $[c - r, c + r]$ — because it makes the cancellation automatic rather than something you have to spot.

1.6 Remark — A Second Proof, Using No Algebra At All

Divisibility by 3 can be settled by looking at residues, and here the residues do all the work.

Proof. Among any three consecutive integers, the three residues mod 3 are all distinct: consecutive integers have consecutive residues, and three steps run once through $\{0, 1, 2\}$. So the three residues are 0, 1, 2 in some order, and

$$0 + 1 + 2 = 3 \equiv 0 \pmod{3}.$$

Hence the sum is $\equiv 0 \pmod{3}$. ■

Verified: for every k from 2 to 12 and every starting point, k consecutive integers do hit each residue mod k exactly once.

The two proofs are genuinely different in flavour. The algebraic one produces the *quotient* — the sum is $3n$, so dividing by 3 returns the middle integer. The residue one produces only the divisibility, but generalises more transparently, as the next remark shows.

1.7 Remark — What Happens for k Consecutive Integers

The natural generalisation is not what one might guess. Summing k consecutive integers starting at n :

$$\sum_{i=0}^{k-1} (n + i) = kn + \frac{k(k-1)}{2}.$$

The first term is divisible by k ; the second is the question. Reducing mod k :

k	sum	sum mod k	k	sum?
2	$2n + 1$	1		no
3	$3n + 3$	0		yes
4	$4n + 6$	2		no
5	$5n + 10$	0		yes
6	$6n + 15$	3		no
7	$7n + 21$	0		yes

So the criterion is parity:

k divides the sum of every k consecutive integers if and only if k is odd.

The reason is visible in $\frac{k(k-1)}{2}$. If k is odd then $\frac{k-1}{2}$ is an integer and the term is $k \cdot \frac{k-1}{2}$, a multiple of k . If k is even then the term is $\frac{k}{2} \cdot (k-1) \equiv \frac{k}{2} \cdot (-1) \equiv \frac{k}{2} \pmod{k}$, which is never 0. Confirmed computationally: for $k = 2, 4, 6, 8$ the sum leaves remainder exactly 1, 2, 3, 4 respectively, always.

The residue proof gives the same criterion at once: the residues sum to $0 + 1 + \cdots + (k-1) = \frac{k(k-1)}{2}$, and that is $\equiv 0 \pmod{k}$ precisely when k is odd. Equivalently — and this is perhaps the cleanest way to say it — the average of k consecutive integers is the middle one when k is odd, and a half-integer when k is even.

1.8 Remark — Sums Versus Products

It is worth setting this beside the corresponding statement for products, because the two behave oppositely.

sum of k consecutive integers	divisible by k exactly when k is odd
product of k consecutive integers	divisible by $k!$ for every k

The product statement is far stronger — divisibility by $k!$ rather than merely k — and holds without exception, because $\frac{n(n+1) \cdots (n+k-1)}{k!} = \binom{n+k-1}{k}$ is a binomial coefficient, hence an integer. Verified for $k \leq 7$ over $1 \leq n < 60$.

For $k = 3$ the two statements read: the sum of three consecutive integers is divisible by 3, and their product is divisible by $3! = 6$. The second is the fact used in Problem 1 of this section, where $m(m+1)$ being even was the crux of the lemma — the $k = 2$ case of the product statement.

So consecutive integers are a small machine for manufacturing divisibility, and it is worth knowing which handle produces which guarantee.